Day 28 — Real-Time Booking Project : Create Spark SQL reporting output.

COMMANDS :

```
cat > src/main/scala/Day28Booking.scala << 'EOF'
import org.apache.spark.SparkConf
import org.apache.spark.broadcast.Broadcast
import org.apache.spark.sql.SparkSession
import org.apache.spark.streaming.{Seconds, StreamingContext}
import scala.util.Try

// ---------- Event schema ----------
case class BookingEvent(eventId: Long, bookingId: String, category: String,
                        resourceId: String, action: String, units: Int)

object BookingEvent {
  private val Categories = Set("HOTEL", "FLIGHT", "BUS")
  def parse(line: String): Option[BookingEvent] = {
    val a = line.trim.split(",").map(_.trim)
    if (a.length != 6) None
    else Try(BookingEvent(a(0).toLong, a(1).toUpperCase, a(2).toUpperCase,
                          a(3).toUpperCase, a(4).toUpperCase, a(5).toInt)).toOption
      .filter(e => e.units > 0 && Categories.contains(e.category) &&
                   (e.action == "BOOK" || e.action == "CANCEL"))
  }
}

// Per-booking state kept across batches
case class BookingState(resourceId: String, units: Int, active: Boolean)

// Rows for Spark SQL
case class OccupancyRow(resourceId: String, name: String, category: String,
                        capacity: Int, booked: Int, available: Int, occupancyPct: Double)
case class ActivityRow(resourceId: String, bookings: Int, cancellations: Int)

object Day28Booking {
  def main(args: Array[String]): Unit = {
    val conf = new SparkConf().setAppName("Day28Booking")
    if (!conf.contains("spark.master")) conf.setMaster("local[2]")

    val ssc = new StreamingContext(conf, Seconds(5))
    val sc  = ssc.sparkContext
    sc.setLogLevel("WARN")
    ssc.checkpoint("/tmp/day28-checkpoint")   // needed for updateStateByKey
```

```
    // ---------- Broadcast reference data: resourceId -> (name, category, capacity) ----------
    val ref: Broadcast[Map[String, (String, String, Int)]] = sc.broadcast(Map(
      "H101" -> ("Taj-Grand",     "HOTEL",  5),
      "H102" -> ("Lakeview-Inn",  "HOTEL",  3),
      "F201" -> ("AI-501",        "FLIGHT", 4),
      "B301" -> ("Volvo-HYD-BLR", "BUS",    4)
    ))

    // ---------- Process booking events ----------
    // Drop malformed lines, unknown resources, and category mismatches (checked against the
broadcast)
    val events = ssc.socketTextStream("localhost", 9999)
      .filter(_.trim.nonEmpty)
      .flatMap(l => BookingEvent.parse(l).toList)
      .filter(e => ref.value.get(e.resourceId).exists(_._2 == e.category))

    // ---------- Booking/cancellation STATE, one entry per bookingId (Pair DStream) ----------
    val updateFunc = (evs: Seq[BookingEvent], st: Option[BookingState]) => {
      var cur = st
      // arrival order inside a batch is not guaranteed, so apply events in eventId order
      evs.sortBy(_.eventId).foreach { e =>
        if (e.action == "BOOK" && cur.isEmpty)
          cur = Some(BookingState(e.resourceId, e.units, active = true))
        else if (e.action == "CANCEL" && cur.exists(_.active))
          cur = cur.map(_.copy(active = false))
        // duplicate BOOK, or CANCEL for an unknown/already-cancelled booking: ignored
      }
      cur
    }
    val bookingState = events.map(e => (e.bookingId,
e)).updateStateByKey[BookingState](updateFunc)

    // ---------- Occupancy: active booked units per resource (Pair RDD reduceByKey) ----------
    val bookedByResource = bookingState
      .map { case (_, s) => (s.resourceId, if (s.active) s.units else 0) }
      .reduceByKey(_ + _)

    // ---------- Spark SQL report: occupancy and availability ----------
    bookedByResource.foreachRDD { (rdd, time) =>
      val booked = rdd.collect().toMap
      val rows = ref.value.toSeq.sortBy(_._1).map { case (rid, (name, cat, cap)) =>
        val b = booked.getOrElse(rid, 0)
        OccupancyRow(rid, name, cat, cap, b, cap - b, math.round(1000.0 * b / cap) / 10.0)
```

```
    }
    val spark = SparkSession.builder().config(rdd.sparkContext.getConf).getOrCreate()
    spark.createDataFrame(rows).createOrReplaceTempView("occupancy")

    println(s"\n===== OCCUPANCY REPORT ($time) =====")
    spark.sql(
      """SELECT category, resourceId, name, capacity, booked, available, occupancyPct,
        |       CASE WHEN available < 0 THEN 'OVERBOOKED'
        |            WHEN available = 0 THEN 'FULL'
        |            WHEN occupancyPct >= 80 THEN 'ALMOST FULL'
        |            ELSE 'OK' END AS status
        |FROM occupancy ORDER BY category, resourceId""".stripMargin).show(false)
    spark.sql(
      """SELECT category, SUM(capacity) AS capacity, SUM(booked) AS booked,
        |       SUM(available) AS available,
        |       ROUND(100.0 * SUM(booked) / SUM(capacity), 1) AS occupancyPct
        |FROM occupancy GROUP BY category ORDER BY category""".stripMargin).show(false)
  }

  // ---------- WINDOW: bookings vs cancellations in the last 30s (slide 10s) ----------
  val activity = events
    .map(e => (e.resourceId, (if (e.action == "BOOK") 1 else 0, if (e.action == "CANCEL") 1 else
0)))
    .reduceByKeyAndWindow((a: (Int, Int), b: (Int, Int)) => (a._1 + b._1, a._2 + b._2),
                          Seconds(30), Seconds(10))

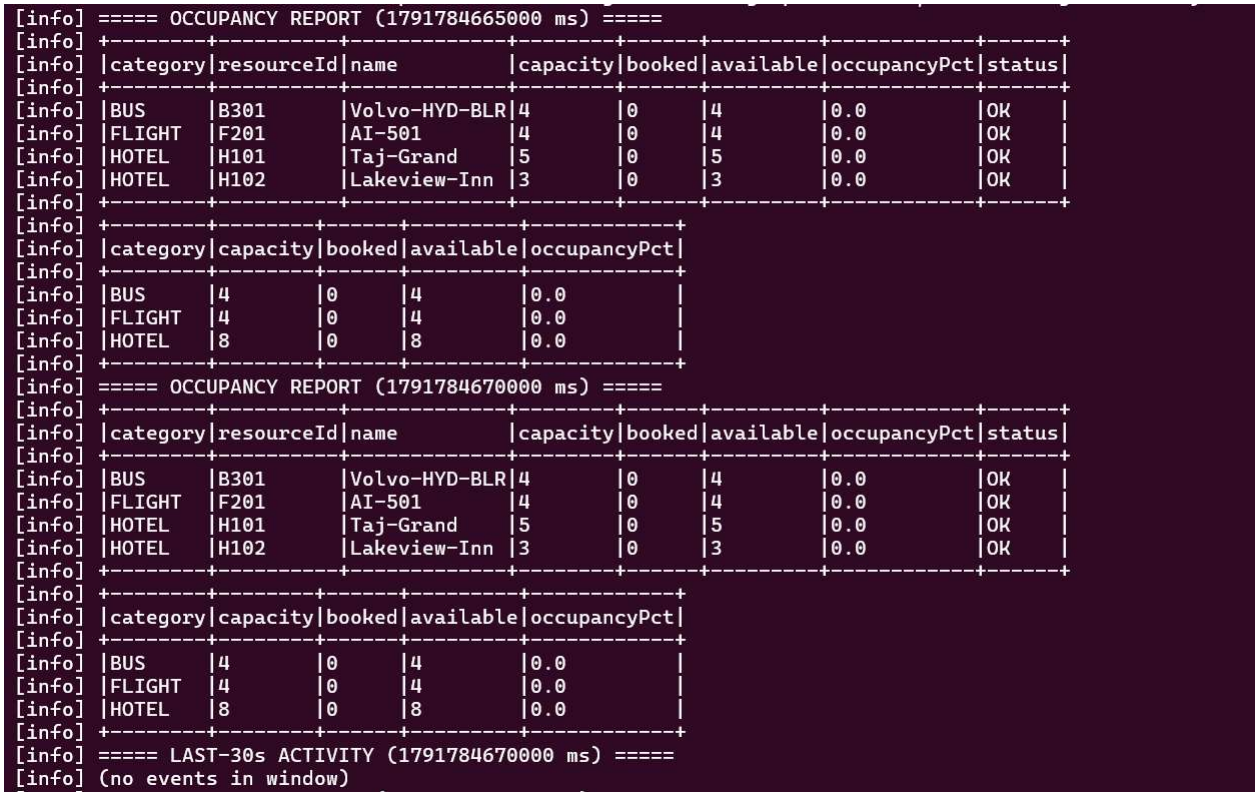
  activity.foreachRDD { (rdd, time) =>
    val rows = rdd.collect().toSeq.map { case (rid, (b, c)) => ActivityRow(rid, b, c) }
    println(s"\n===== LAST-30s ACTIVITY ($time) =====")
    if (rows.isEmpty) println("(no events in window)")
    else {
      val spark = SparkSession.builder().config(rdd.sparkContext.getConf).getOrCreate()
      spark.createDataFrame(rows).createOrReplaceTempView("activity")
      spark.sql(
        """SELECT resourceId, bookings, cancellations,
          |       COALESCE(ROUND(100.0 * cancellations / NULLIF(bookings, 0), 1), 0.0) AS
cancelRatePct
          |FROM activity ORDER BY bookings DESC, resourceId""".stripMargin).show(false)
    }
  }

  ssc.start()
  println("Day 28 started. Port 9999
(EVENTID,BOOKINGID,CATEGORY,RESOURCE,ACTION,UNITS)...")
```

```
    ssc.awaitTermination()
  }
}
EOF

cd ~/day23-dstreams
rm -rf /tmp/day28-checkpoint
sbt "runMain Day28Booking"
```

OUTPUTS :



```
[info] ===== OCCUPANCY REPORT (1791784665000 ms) =====
[info] +--------+----------+-------------+--------+------+---------+------------+------+
[info] |category|resourceId|name         |capacity|booked|available|occupancyPct|status|
[info] +--------+----------+-------------+--------+------+---------+------------+------+
[info] |BUS     |B301      |Volvo-HYD-BLR|4       |0     |4        |0.0         |OK    |
[info] |FLIGHT  |F201      |AI-501       |4       |0     |4        |0.0         |OK    |
[info] |HOTEL   |H101      |Taj-Grand    |5       |0     |5        |0.0         |OK    |
[info] |HOTEL   |H102      |Lakeview-Inn |3       |0     |3        |0.0         |OK    |
[info] +--------+----------+-------------+--------+------+---------+------------+------+
[info] +--------+--------+------+---------+------------+
[info] |category|capacity|booked|available|occupancyPct|
[info] +--------+--------+------+---------+------------+
[info] |BUS     |4       |0     |4        |0.0         |
[info] |FLIGHT  |4       |0     |4        |0.0         |
[info] |HOTEL   |8       |0     |8        |0.0         |
[info] +--------+--------+------+---------+------------+
[info] ===== OCCUPANCY REPORT (1791784670000 ms) =====
[info] +--------+----------+-------------+--------+------+---------+------------+------+
[info] |category|resourceId|name         |capacity|booked|available|occupancyPct|status|
[info] +--------+----------+-------------+--------+------+---------+------------+------+
[info] |BUS     |B301      |Volvo-HYD-BLR|4       |0     |4        |0.0         |OK    |
[info] |FLIGHT  |F201      |AI-501       |4       |0     |4        |0.0         |OK    |
[info] |HOTEL   |H101      |Taj-Grand    |5       |0     |5        |0.0         |OK    |
[info] |HOTEL   |H102      |Lakeview-Inn |3       |0     |3        |0.0         |OK    |
[info] +--------+----------+-------------+--------+------+---------+------------+------+
[info] +--------+--------+------+---------+------------+
[info] |category|capacity|booked|available|occupancyPct|
[info] +--------+--------+------+---------+------------+
[info] |BUS     |4       |0     |4        |0.0         |
[info] |FLIGHT  |4       |0     |4        |0.0         |
[info] |HOTEL   |8       |0     |8        |0.0         |
[info] +--------+--------+------+---------+------------+
[info] ===== LAST-30s ACTIVITY (1791784670000 ms) =====
[info] (no events in window)
```